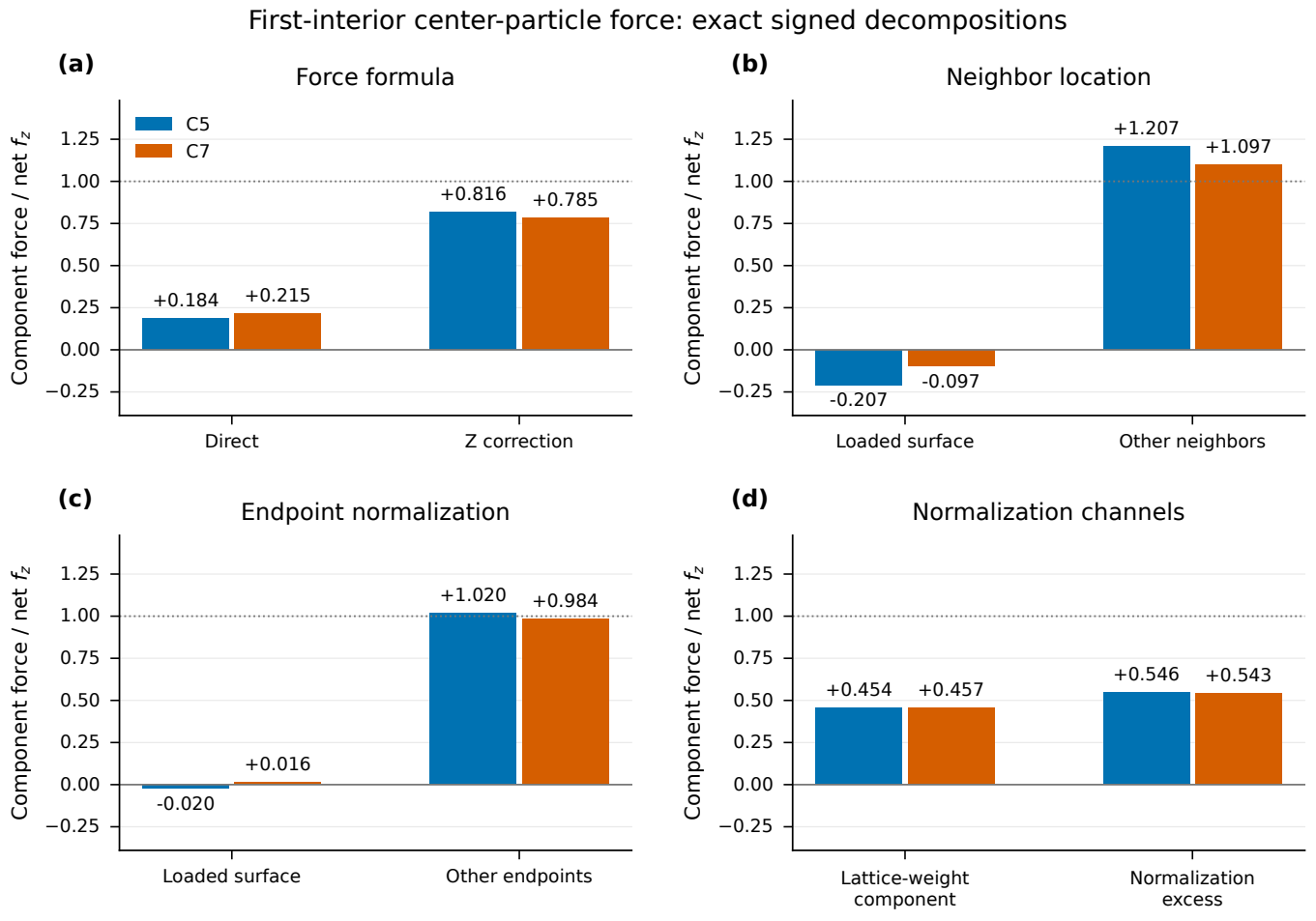
First-interior center-particle force: exact signed decompositions
(a)
Force formula
C5
C7
Component force / net f_z
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
+0.184
+0.215
+0.816
+0.785
Direct
Z correction
(b)
Neighbor location
Component force / net f_z
+1.207
+1.097
-0.207
-0.097
Loaded surface
Other neighbors
(c)
Endpoint normalization
Component force / net f_z
+1.020
+0.984
-0.020
+0.016
Loaded surface
Other endpoints
(d)
Normalization channels
Component force / net f_z
+0.454
+0.457
+0.546
+0.543
Lattice-weight component
Normalization excess